

WHAM-XREX-PFMG474 Closed-Loop Gain Smoothness Sweep

Generated by `python/run_simulator_validation.py`

September 22, 2026

Purpose

5 Kp values (0.0, 0.1, 0.2, 0.3, 0.5) x 4 Ki values (250.0, 750.0, 1500.0, 3000.0), Kd=0 throughout, channel 1 only at a fixed 3000 A demand, each a real 3s/3s/3s ramp/flat-top/ramp-down profile (9s total). Motivated by the 2026-09-21 45-shot matrix campaign (Kp=0.5, Ki=750): every shot converged, but the commanded OUTPUT (drive) trace was visibly noisier than earlier tests – suspected cause is the PID not acting enough like an integrator (the P term reacts instantly to noisy feedback every sample; the I term inherently smooths it over time). Ranked below by output jitter RMS (root-mean-square of sample-to-sample differences in the commanded output over the flat-top window, 0.5s trimmed off each end) – smoothest first – with tracking convergence reported alongside as a sanity check only, not the ranking criterion. Full raw data for every shot is archived under `shots/` in this repository (gitignored).

Summary

20/20 shots converged to within 1500 Hz (flat-top median tracking error). Ranked by output smoothness below – lower jitter RMS is smoother.

Ranked Results (smoothest first)

Kp	Ki	Output jitter RMS (Hz)	Output std (Hz)	Tracking median error (Hz)	Converged?
0.1	250	781	561	525	PASS
0	250	790	598	513	PASS
0.2	250	847	607	509	PASS
0.3	250	881	614	513	PASS
0.5	250	979	611	513	PASS
0.2	750	2556	2071	525	PASS
0.3	750	2558	2031	558	PASS
0.5	750	2608	1820	558	PASS
0.1	750	2670	2346	509	PASS
0	750	3125	3648	558	PASS
0.5	1500	7301	9540	970	PASS
0.2	1500	7717	10702	1048	PASS

Kp	Ki	Output jitter RMS (Hz)	Output std (Hz)	Tracking median error (Hz)	Converged?
0.1	1500	7751	10545	1077	PASS
0	1500	7889	11052	1048	PASS
0.3	1500	7890	11186	1077	PASS
0.2	3000	8832	11311	959	PASS
0.3	3000	9779	13958	1280	PASS
0.5	3000	9810	13771	1280	PASS
0.1	3000	10064	13977	1416	PASS
0	3000	10216	14377	1434	PASS

Per-Shot Detail (smoothest first)

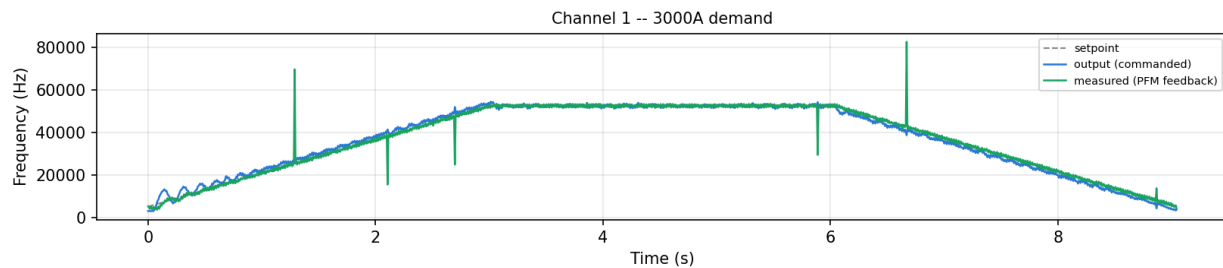
1 Rank 1: $K_p=0.1$, $K_i=250$

Channel 1, closed loop ($K_p=0.1$, $K_i=250$, $K_d=0$), commanding a real 3s/3s/3s ramp/flat-top/ramp-down profile to 3000 A demand. Output jitter RMS 781 Hz, output std 561 Hz over the flat-top window; flat-top median tracking error 525 Hz – converged to setpoint.

Verdict: PASS

1.1 Commanded output vs. measured feedback

$K_p=0.1$, $K_i=250$, $K_d=0$ -- 3000 A demand -- output jitter RMS 781 Hz, tracking median error 525 Hz



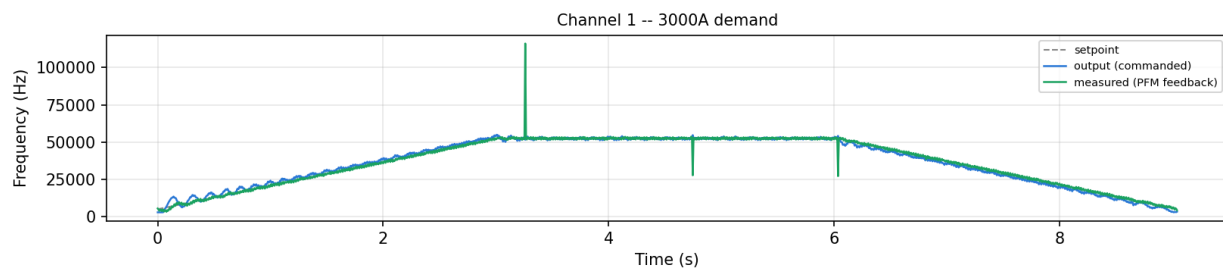
2 Rank 2: $K_p=0$, $K_i=250$

Channel 1, closed loop ($K_p=0$, $K_i=250$, $K_d=0$), commanding a real 3s/3s/3s ramp/flat-top/ramp-down profile to 3000 A demand. Output jitter RMS 790 Hz, output std 598 Hz over the flat-top window; flat-top median tracking error 513 Hz – converged to setpoint.

Verdict: PASS

2.1 Commanded output vs. measured feedback

$K_p=0$, $K_i=250$, $K_d=0$ -- 3000 A demand -- output jitter RMS 790 Hz, tracking median error 513 Hz



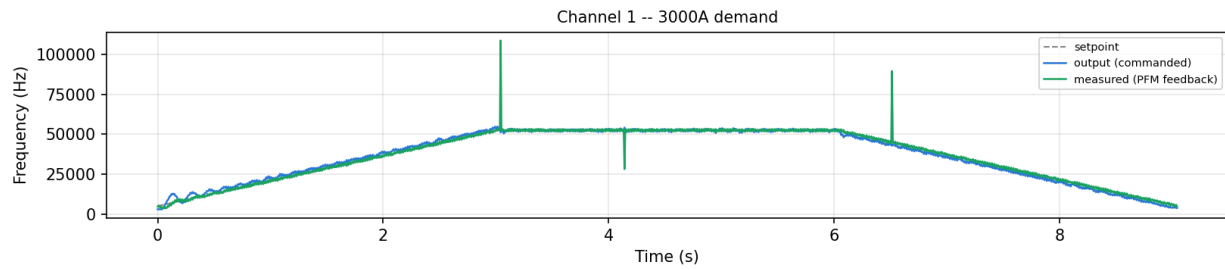
3 Rank 3: $K_p=0.2$, $K_i=250$

Channel 1, closed loop ($K_p=0.2$, $K_i=250$, $K_d=0$), commanding a real 3s/3s/3s ramp/flat-top/ramp-down profile to 3000 A demand. Output jitter RMS 847 Hz, output std 607 Hz over the flat-top window; flat-top median tracking error 509 Hz – converged to setpoint.

Verdict: PASS

3.1 Commanded output vs. measured feedback

$K_p=0.2$, $K_i=250$, $K_d=0$ -- 3000 A demand -- output jitter RMS 847 Hz, tracking median error 509 Hz



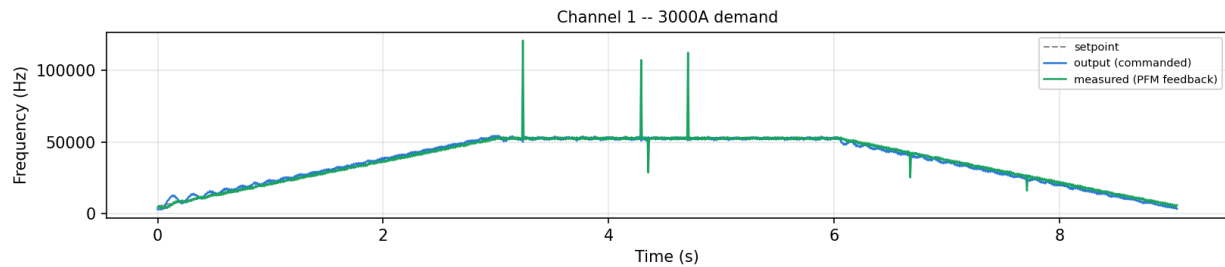
4 Rank 4: $K_p=0.3$, $K_i=250$

Channel 1, closed loop ($K_p=0.3$, $K_i=250$, $K_d=0$), commanding a real 3s/3s/3s ramp/flat-top/ramp-down profile to 3000 A demand. Output jitter RMS 881 Hz, output std 614 Hz over the flat-top window; flat-top median tracking error 513 Hz – converged to setpoint.

Verdict: PASS

4.1 Commanded output vs. measured feedback

$K_p=0.3$, $K_i=250$, $K_d=0$ -- 3000 A demand -- output jitter RMS 881 Hz, tracking median error 513 Hz



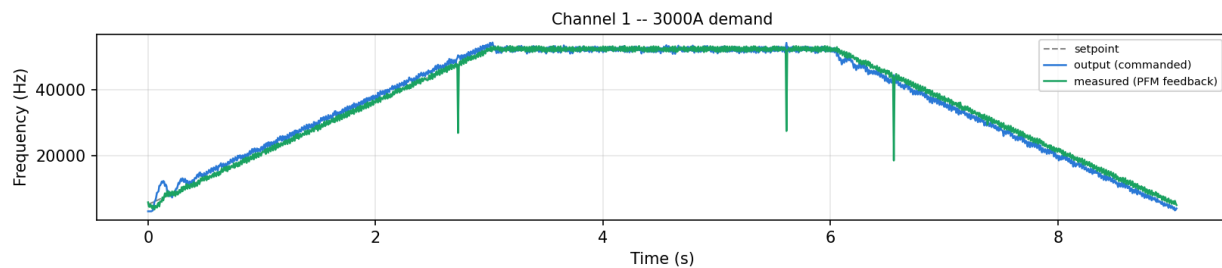
5 Rank 5: $K_p=0.5$, $K_i=250$

Channel 1, closed loop ($K_p=0.5$, $K_i=250$, $K_d=0$), commanding a real 3s/3s/3s ramp/flat-top/ramp-down profile to 3000 A demand. Output jitter RMS 979 Hz, output std 611 Hz over the flat-top window; flat-top median tracking error 513 Hz – converged to setpoint.

Verdict: PASS

5.1 Commanded output vs. measured feedback

$K_p=0.5$, $K_i=250$, $K_d=0$ -- 3000 A demand -- output jitter RMS 979 Hz, tracking median error 513 Hz



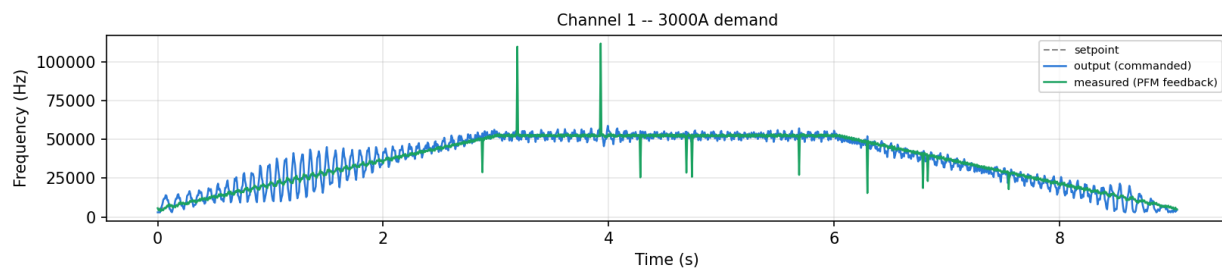
6 Rank 6: $K_p=0.2$, $K_i=750$

Channel 1, closed loop ($K_p=0.2$, $K_i=750$, $K_d=0$), commanding a real 3s/3s/3s ramp/flat-top/ramp-down profile to 3000 A demand. Output jitter RMS 2556 Hz, output std 2071 Hz over the flat-top window; flat-top median tracking error 525 Hz – converged to setpoint.

Verdict: PASS

6.1 Commanded output vs. measured feedback

$K_p=0.2$, $K_i=750$, $K_d=0$ -- 3000 A demand -- output jitter RMS 2556 Hz, tracking median error 525 Hz



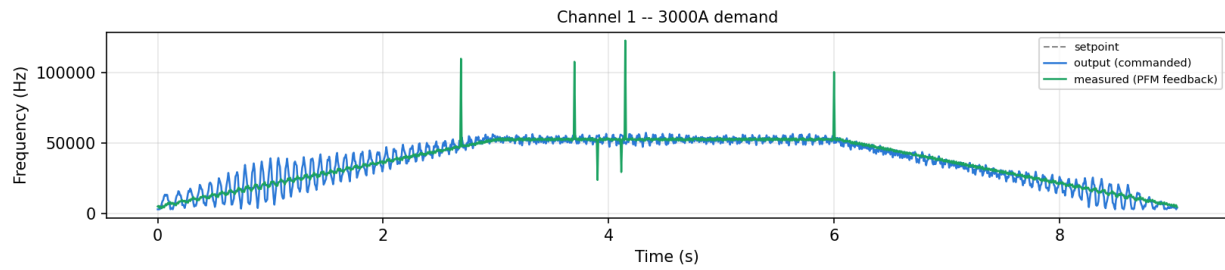
7 Rank 7: $K_p=0.3$, $K_i=750$

Channel 1, closed loop ($K_p=0.3$, $K_i=750$, $K_d=0$), commanding a real 3s/3s/3s ramp/flat-top/ramp-down profile to 3000 A demand. Output jitter RMS 2558 Hz, output std 2031 Hz over the flat-top window; flat-top median tracking error 558 Hz – converged to setpoint.

Verdict: PASS

7.1 Commanded output vs. measured feedback

$K_p=0.3$, $K_i=750$, $K_d=0$ -- 3000 A demand -- output jitter RMS 2558 Hz, tracking median error 558 Hz



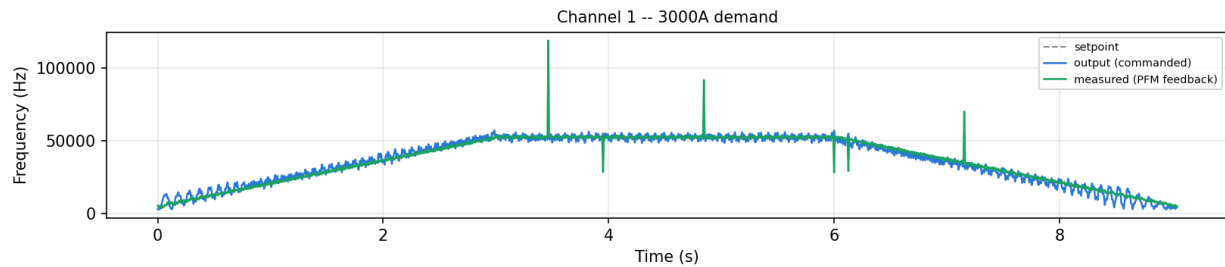
8 Rank 8: $K_p=0.5$, $K_i=750$

Channel 1, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 3s/3s/3s ramp/flat-top/ramp-down profile to 3000 A demand. Output jitter RMS 2608 Hz, output std 1820 Hz over the flat-top window; flat-top median tracking error 558 Hz – converged to setpoint.

Verdict: PASS

8.1 Commanded output vs. measured feedback

$K_p=0.5$, $K_i=750$, $K_d=0$ -- 3000 A demand -- output jitter RMS 2608 Hz, tracking median error 558 Hz



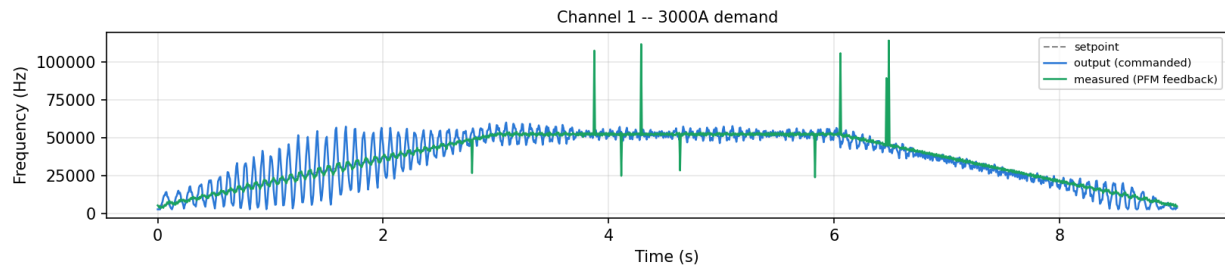
9 Rank 9: $K_p=0.1$, $K_i=750$

Channel 1, closed loop ($K_p=0.1$, $K_i=750$, $K_d=0$), commanding a real 3s/3s/3s ramp/flat-top/ramp-down profile to 3000 A demand. Output jitter RMS 2670 Hz, output std 2346 Hz over the flat-top window; flat-top median tracking error 509 Hz – converged to setpoint.

Verdict: PASS

9.1 Commanded output vs. measured feedback

$K_p=0.1$, $K_i=750$, $K_d=0$ -- 3000 A demand -- output jitter RMS 2670 Hz, tracking median error 509 Hz



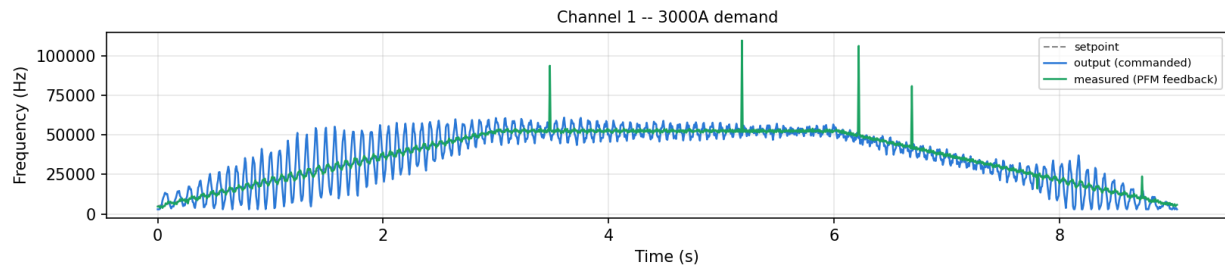
10 Rank 10: $K_p=0$, $K_i=750$

Channel 1, closed loop ($K_p=0$, $K_i=750$, $K_d=0$), commanding a real 3s/3s/3s ramp/flat-top/ramp-down profile to 3000 A demand. Output jitter RMS 3125 Hz, output std 3648 Hz over the flat-top window; flat-top median tracking error 558 Hz – converged to setpoint.

Verdict: PASS

10.1 Commanded output vs. measured feedback

$K_p=0$, $K_i=750$, $K_d=0$ -- 3000 A demand -- output jitter RMS 3125 Hz, tracking median error 558 Hz



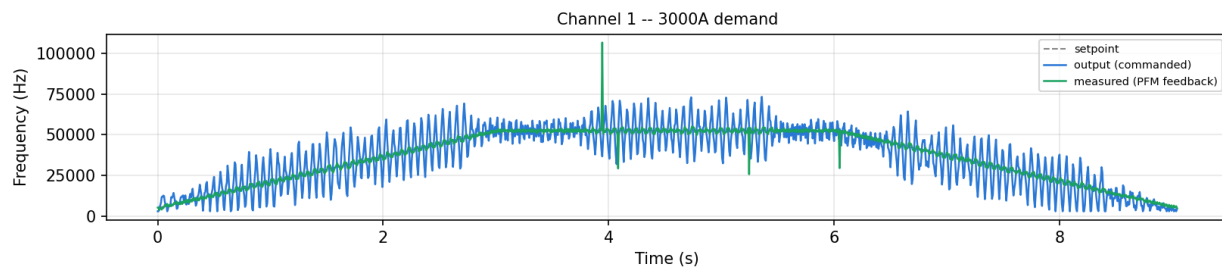
11 Rank 11: $K_p=0.5$, $K_i=1500$

Channel 1, closed loop ($K_p=0.5$, $K_i=1500$, $K_d=0$), commanding a real 3s/3s/3s ramp/flat-top/ramp-down profile to 3000 A demand. Output jitter RMS 7301 Hz, output std 9540 Hz over the flat-top window; flat-top median tracking error 970 Hz – converged to setpoint.

Verdict: PASS

11.1 Commanded output vs. measured feedback

$K_p=0.5$, $K_i=1500$, $K_d=0$ -- 3000 A demand -- output jitter RMS 7301 Hz, tracking median error 970 Hz



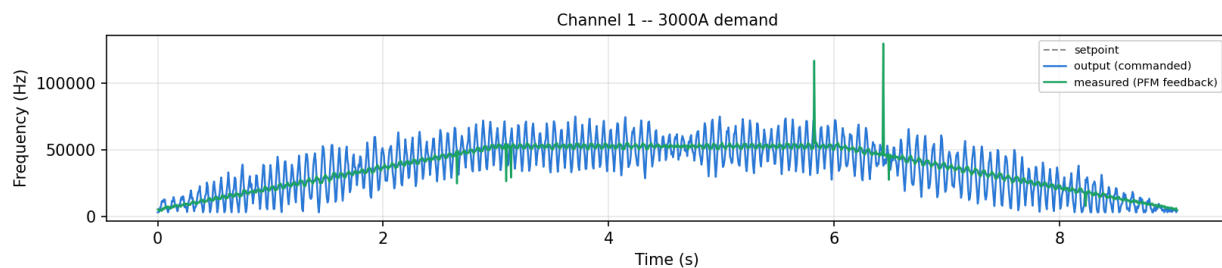
12 Rank 12: $K_p=0.2$, $K_i=1500$

Channel 1, closed loop ($K_p=0.2$, $K_i=1500$, $K_d=0$), commanding a real 3s/3s/3s ramp/flat-top/ramp-down profile to 3000 A demand. Output jitter RMS 7717 Hz, output std 10702 Hz over the flat-top window; flat-top median tracking error 1048 Hz – converged to setpoint.

Verdict: PASS

12.1 Commanded output vs. measured feedback

$K_p=0.2$, $K_i=1500$, $K_d=0$ -- 3000 A demand -- output jitter RMS 7717 Hz, tracking median error 1048 Hz



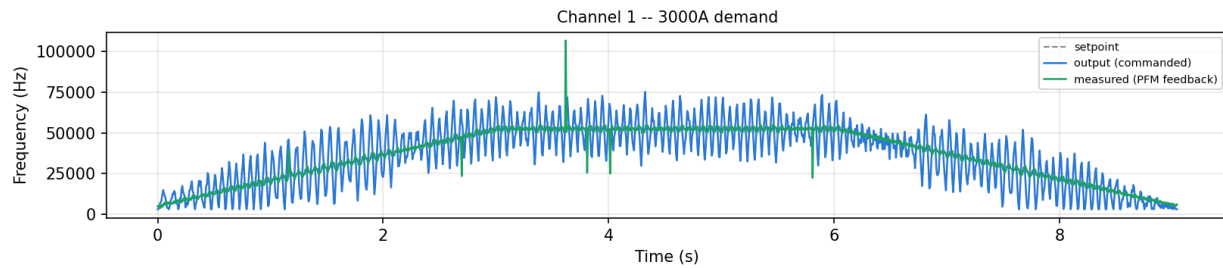
13 Rank 13: $K_p=0.1$, $K_i=1500$

Channel 1, closed loop ($K_p=0.1$, $K_i=1500$, $K_d=0$), commanding a real 3s/3s/3s ramp/flat-top/ramp-down profile to 3000 A demand. Output jitter RMS 7751 Hz, output std 10545 Hz over the flat-top window; flat-top median tracking error 1077 Hz – converged to setpoint.

Verdict: PASS

13.1 Commanded output vs. measured feedback

$K_p=0.1$, $K_i=1500$, $K_d=0$ -- 3000 A demand -- output jitter RMS 7751 Hz, tracking median error 1077 Hz



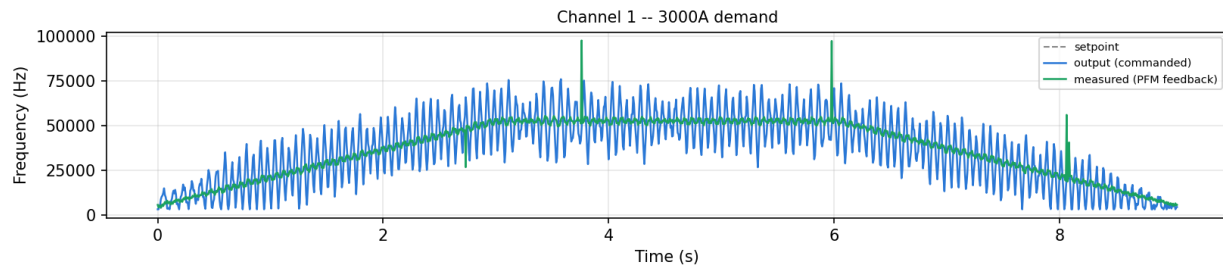
14 Rank 14: $K_p=0$, $K_i=1500$

Channel 1, closed loop ($K_p=0$, $K_i=1500$, $K_d=0$), commanding a real 3s/3s/3s ramp/flat-top/ramp-down profile to 3000 A demand. Output jitter RMS 7889 Hz, output std 11052 Hz over the flat-top window; flat-top median tracking error 1048 Hz – converged to setpoint.

Verdict: PASS

14.1 Commanded output vs. measured feedback

$K_p=0$, $K_i=1500$, $K_d=0$ -- 3000 A demand -- output jitter RMS 7889 Hz, tracking median error 1048 Hz



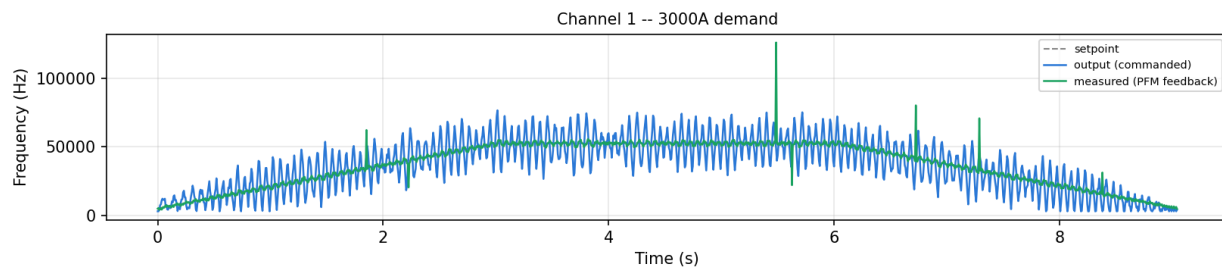
15 Rank 15: $K_p=0.3$, $K_i=1500$

Channel 1, closed loop ($K_p=0.3$, $K_i=1500$, $K_d=0$), commanding a real 3s/3s/3s ramp/flat-top/ramp-down profile to 3000 A demand. Output jitter RMS 7890 Hz, output std 11186 Hz over the flat-top window; flat-top median tracking error 1077 Hz – converged to setpoint.

Verdict: PASS

15.1 Commanded output vs. measured feedback

$K_p=0.3$, $K_i=1500$, $K_d=0$ -- 3000 A demand -- output jitter RMS 7890 Hz, tracking median error 1077 Hz



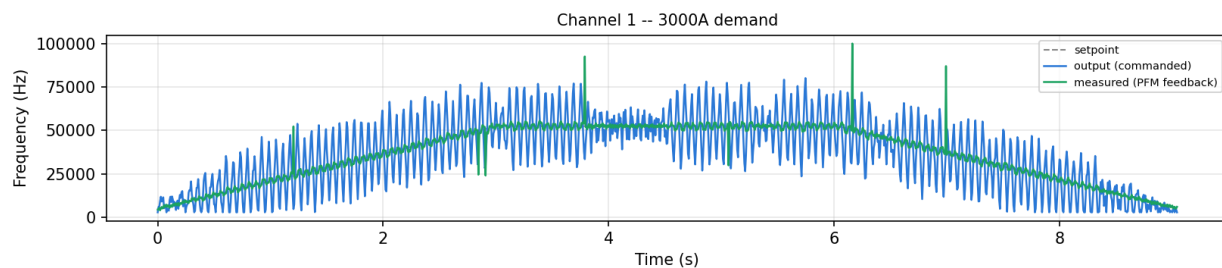
16 Rank 16: $K_p=0.2$, $K_i=3000$

Channel 1, closed loop ($K_p=0.2$, $K_i=3000$, $K_d=0$), commanding a real 3s/3s/3s ramp/flat-top/ramp-down profile to 3000 A demand. Output jitter RMS 8832 Hz, output std 11311 Hz over the flat-top window; flat-top median tracking error 959 Hz – converged to setpoint.

Verdict: PASS

16.1 Commanded output vs. measured feedback

$K_p=0.2$, $K_i=3000$, $K_d=0$ -- 3000 A demand -- output jitter RMS 8832 Hz, tracking median error 959 Hz



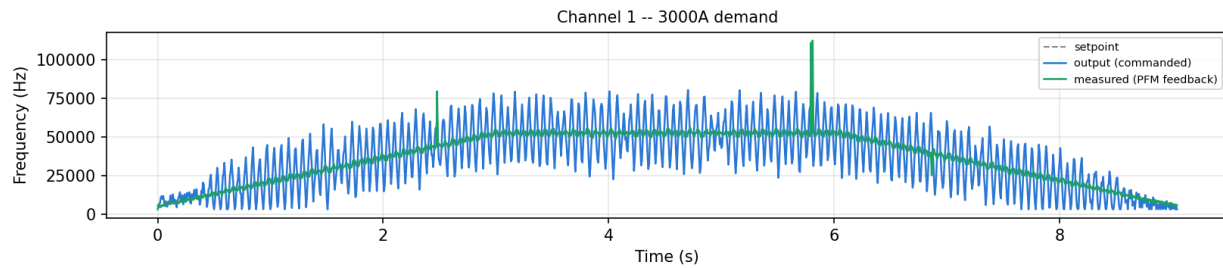
17 Rank 17: $K_p=0.3$, $K_i=3000$

Channel 1, closed loop ($K_p=0.3$, $K_i=3000$, $K_d=0$), commanding a real 3s/3s/3s ramp/flat-top/ramp-down profile to 3000 A demand. Output jitter RMS 9779 Hz, output std 13958 Hz over the flat-top window; flat-top median tracking error 1280 Hz – converged to setpoint.

Verdict: PASS

17.1 Commanded output vs. measured feedback

$K_p=0.3$, $K_i=3000$, $K_d=0$ -- 3000 A demand -- output jitter RMS 9779 Hz, tracking median error 1280 Hz



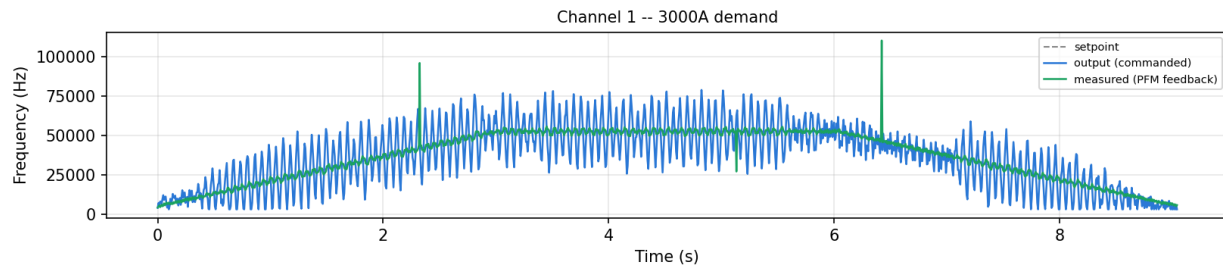
18 Rank 18: $K_p=0.5$, $K_i=3000$

Channel 1, closed loop ($K_p=0.5$, $K_i=3000$, $K_d=0$), commanding a real 3s/3s/3s ramp/flat-top/ramp-down profile to 3000 A demand. Output jitter RMS 9810 Hz, output std 13771 Hz over the flat-top window; flat-top median tracking error 1280 Hz – converged to setpoint.

Verdict: PASS

18.1 Commanded output vs. measured feedback

$K_p=0.5$, $K_i=3000$, $K_d=0$ -- 3000 A demand -- output jitter RMS 9810 Hz, tracking median error 1280 Hz



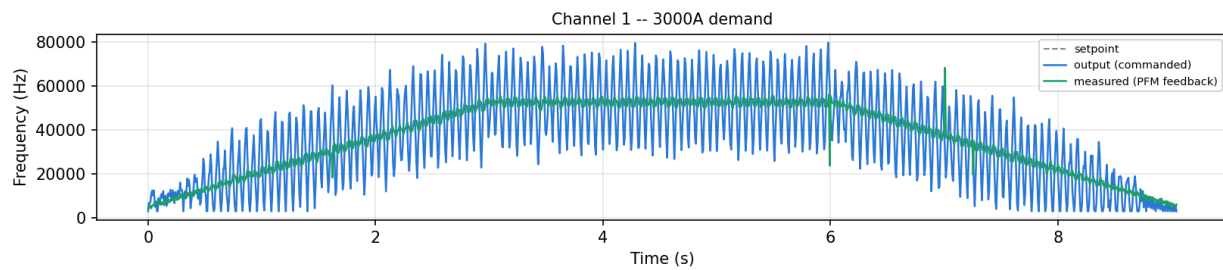
19 Rank 19: $K_p=0.1$, $K_i=3000$

Channel 1, closed loop ($K_p=0.1$, $K_i=3000$, $K_d=0$), commanding a real 3s/3s/3s ramp/flat-top/ramp-down profile to 3000 A demand. Output jitter RMS 10064 Hz, output std 13977 Hz over the flat-top window; flat-top median tracking error 1416 Hz – converged to setpoint.

Verdict: PASS

19.1 Commanded output vs. measured feedback

$K_p=0.1$, $K_i=3000$, $K_d=0$ -- 3000 A demand -- output jitter RMS 10064 Hz, tracking median error 1416 Hz



20 Rank 20: $K_p=0$, $K_i=3000$

Channel 1, closed loop ($K_p=0$, $K_i=3000$, $K_d=0$), commanding a real 3s/3s/3s ramp/flat-top/ramp-down profile to 3000 A demand. Output jitter RMS 10216 Hz, output std 14377 Hz over the flat-top window; flat-top median tracking error 1434 Hz – converged to setpoint.

Verdict: PASS

20.1 Commanded output vs. measured feedback

$K_p=0$, $K_i=3000$, $K_d=0$ -- 3000 A demand -- output jitter RMS 10216 Hz, tracking median error 1434 Hz

